



DEALER BEST PRACTICE SERIES

Work Area Optimization (5S)

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Work Area Optimization (5S)0

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1.0 Introduction

Efficient Component Rebuild Center (CRC) space management is an important factor of productivity. Missing tools in a specific area can cause major time loss and can lead to potential accidents using “will-do” tools instead of the one fit for the job. Overstocking of unnecessary material and tooling increases the cost of the department without adding any value.

Many CRC shops around the globe have been in place for several years. This Best Practice is a structured approach to address facility revamp. The principles outlined can apply to numerous facilities, both for dealers and customers.

2.0 Best Practice Description

Different principles are used in the industry to help organizations manage their working environments. This Best Practice utilizes a workplace organization methodology that originated from Henry Ford, and was later improved by the Japanese. It is commonly known as “**5S**” because it uses a list of five Japanese words, when translated into English, start with the letter S.

1 – **Seiri** = Sort: Go through all the tools, materials, etc. in the work area and keep only essential items. Everything else is stored, returned or discarded.

2 – **Seiton** = Straighten or Set in Order: Focus on efficiency. Arrange the tools, equipment and parts to promote work flow and maximize efficiency. A place for everything and everything is in its place (separation and labeling). Limit the objects sitting on the floor. If possible, make things mobile. Final set-up must take consideration of ergonomics for the workers.

3 – **Seisō** = Sweeping, Shining or Cleanliness: Systematic Cleaning. At the end of each shift, the work area is cleaned up and everything is restored to its place. This makes it easy to know everything is accounted for and where it should be. Everyone becomes a janitor.

4 – **Seiketsu** = Standardizing: Standardized work practices or operating in a consistent and standardized fashion. Everyone knows exactly what his/her responsibilities are to keep above 3S's.

5 – **Shitsuke** = Sustaining the discipline: Maintaining and reviewing standards. Maintain the focus on this new way of operating, and do not allow a gradual decline back to the old ways. However, when an issue arises such as a suggested improvement, a new way of working or a new tool, then a review of the first 4S's is appropriate.

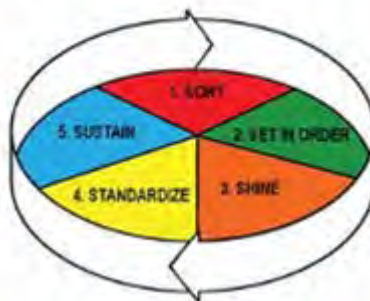


Figure 1 - 5S diagram

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A sixth phase, "Safety," is sometimes added. Purists, however, argue that adding it is unnecessary since following 5S correctly will result in a safe work environment.

Two basic rules of the 5S methodology are as follows:

- It should take no more than 5 seconds to find the tool you need
- It should take no more than 5 minutes to clean the area.

3.0 Implementation Steps

- Assign a team to perform 5S
- Conduct training on the 5S methodology
- Select an area to implement 5S



Figure 2 - Implementing 5S

Because it's not always feasible to halt operations in order to implement 5S, focusing on one area at a time while it's temporarily relocated is advisable in most applications.

1. **Sort:** Create an inventory and discard everything unnecessary



Figure 3 - Sort

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2. **Straighten:** Define a place for everything and put everything in its place

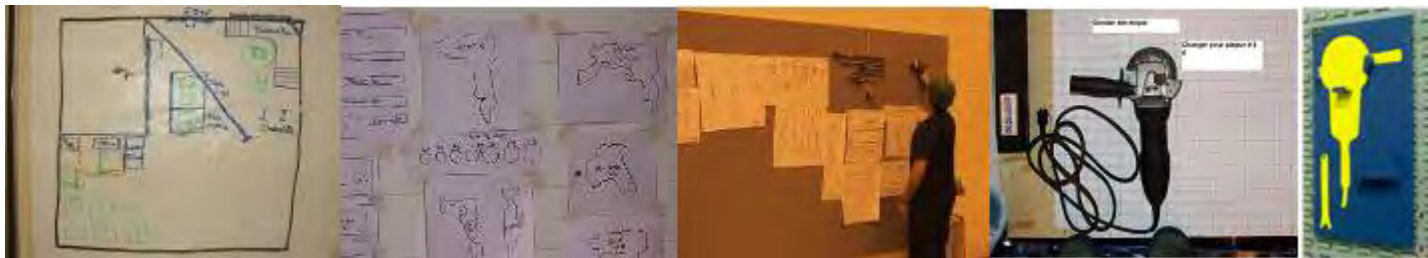


Figure 4 - Straighten

3. **Shine:** Systematic cleaning, painting



Figure 5 - Shine

4. **Standardize**

Color code all tools, identify staging areas for components, ensure parts delivery location, clearly identify rebuilt components. Rebuild procedures should be constantly updated and a copy kept in the designated rebuild area.



Figure 6 - Standardize

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5. Sustain the discipline

Upon completion of implementing 5S, employees will be proud of their work bays and keep them clean, ensuring others do as well. But in time what was once new will become routine. Therefore, an audit schedule must be an integral part of every project with no exception tolerated, constantly work on keeping everyone's commitment alive. Additionally, a maintenance schedule for the area should be defined (painting, equipment maintenance, contamination control and process audits, etc.).

4.0 Benefits

- Additional work bays utilizing the same floor plan
- Improvements in safety, productivity and engagement
- Eliminate excess parts and tools
- Tools become more accessible and are more easily accounted for
- Consistent workflow and quality

At a customer's site, the implementation should include the dealer's participation to make sure that the necessary tooling and procedures are correct, and that contamination control included.



Figure 7 - CRC before and after 5S implementation

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5.0 Resources Required

- Project sponsor
- Logistic manager
- Trainer (1 day)
- 2 full time employees
- 1 painter (occasional)
- 1 welder to fabricate catwalks and mobile carts (occasional)
- 1 drawing engineer for above (occasional)
- Dealer involvement for contamination control, special tooling and procedures advice.
- Project duration per area: approx 8-10 days w/ 30 min. meetings beginning & end of each day
- Shelving, racking, High density polyethylene (HDPE) for assembly tables, labeling, paint etc...

6.0 Supporting Attachments / References



Figure 8 - Mobile cart with an oil retention tub, designed and built in house for final drives



Figure 9 - Hydraulic adjustable height tables with HDPE liner for component assembly

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7.0 Related Best Practices

0107-4.4-1057 *CRC Hand Tool Management*

0906-4.5-1013 *Improving CRC Rebuild Turnaround Time*

8.0 Acknowledgements

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